

Chapter 1

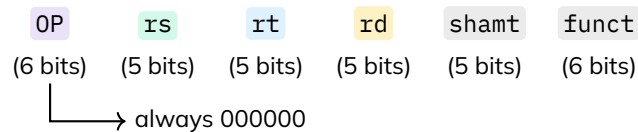
MIPS

1.1 MIPS Instructions

These are the three main instruction formats in the MIPS (Microprocessor without Interlocked Pipeline Stages) architecture: R, I, J

1.1.1 R - Register Type

R-type instructions are used for operations that involve only registers. They have three register operands and a function code to specify the operation.



Example: add \$t0, \$t1, \$t2 → (Addition) \$t0 = \$t1 + \$t2
sll \$t0, \$t1, 2 → (Shift Left) \$t0 = \$t1 << 2

1.1.2 I - Immediate Type

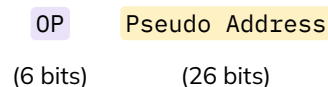
I-type instructions are used for operations that involve an immediate value (a constant) and registers. They have two register operands and a 16-bit immediate value.



Example: lw \$t0, 4(\$t1) → Load word from memory at address (\$t1+4) into \$t0

1.1.3 J - Jump Type

J-type instructions are used for jump operations, which change the program counter to a new address. They have a 26-bit target address.



Example: j TARGET → Jump to the instruction at address TARGET
jal FUNCTION → Jump to FUNCTION and save return address in \$ra

1.2 Common MIPS Instruction Set

1.2.1 Arithmetic Instructions

- add - Add two registers
- addi - Add reg and immediate
- addu - unsinged
- sub - Subtract two registers
- subi - Sub reg and immediate
- subu - sub unsigned
- mul - Multiply two registers
- div - Divide two registers

1.2.2 Logical Instructions

- and - Bitwise AND
- or - Bitwise OR
- xor - Bitwise XOR
- nor - Bitwise NOR
- not - Bitwise Not

1.2.3 Shift Instructions

- sll - Shift Left Logical
- srl - Shift Right Logical
- sra - Shift Right Arithmetic

1.2.4 Memory Access Instructions

- lw - Load Word (from memory to a register)
- sw - Store Word (from a register to memory)
- lb - Load Byte (from memory to a register)
- sb - Store Byte (from a register to memory)

1.2.5 Control Flow Instructions

- beq - Branch if Equal
- bne - Branch if Not Equal
- j - Jump to a specified address
- jal - Jump and Link (used for function calls)
- jr - Jump Register (used for returning from a function)

1.3 MIPS Registers

Register	#Reg no.	Name	Usage	Preserved across calls?
\$zero	0	Zero Register	Always holds 0	Yes
\$at	1	Assembler Temp	Reserved for assembler (pseudo-instructions)	No
\$v0-\$v1	2-3	Return Values	Used for function return values	No
\$a0-\$a3	4-7	Arguments	Used to pass function arguments	No
\$t0-\$t7	8-15	Temporary	Used for temporary values	No
\$s0-\$s7	16-23	Saved Registers	Used for saved values, preserved across calls	Yes
\$t8-\$t9	24-25	Temporary	Additional temporary registers	No
\$k0-\$k1	26-27	Kernel Reserved	Used by the OS kernel	N/A
\$gp	28	Global Pointer	Points to global variables	Yes
\$sp	29	Stack Pointer	Points to the top of the stack	Yes
\$fp	30	Frame Pointer	Used for function stack frames	Yes
\$ra	31	Return Address	Stores return address for function calls	No

Table 1.1: MIPS Register Table